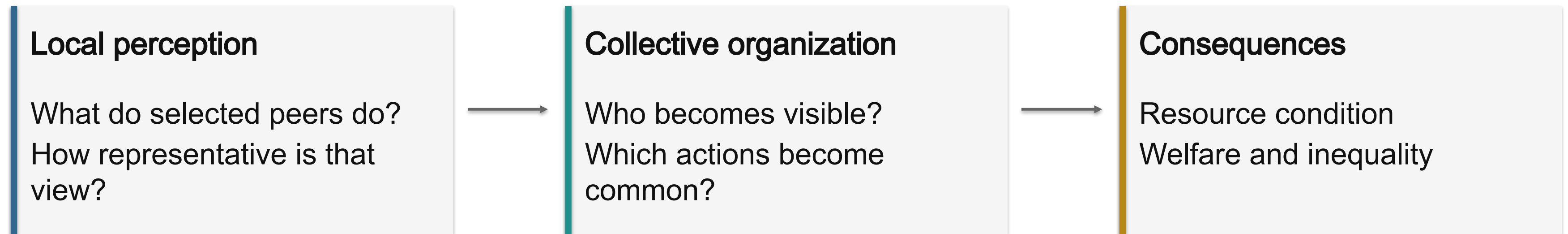


How does ecology shape perception and collective organization?

RQ1: When do local observations misrepresent population behavior?

RQ2: How do ecology and observation rules organize the population?



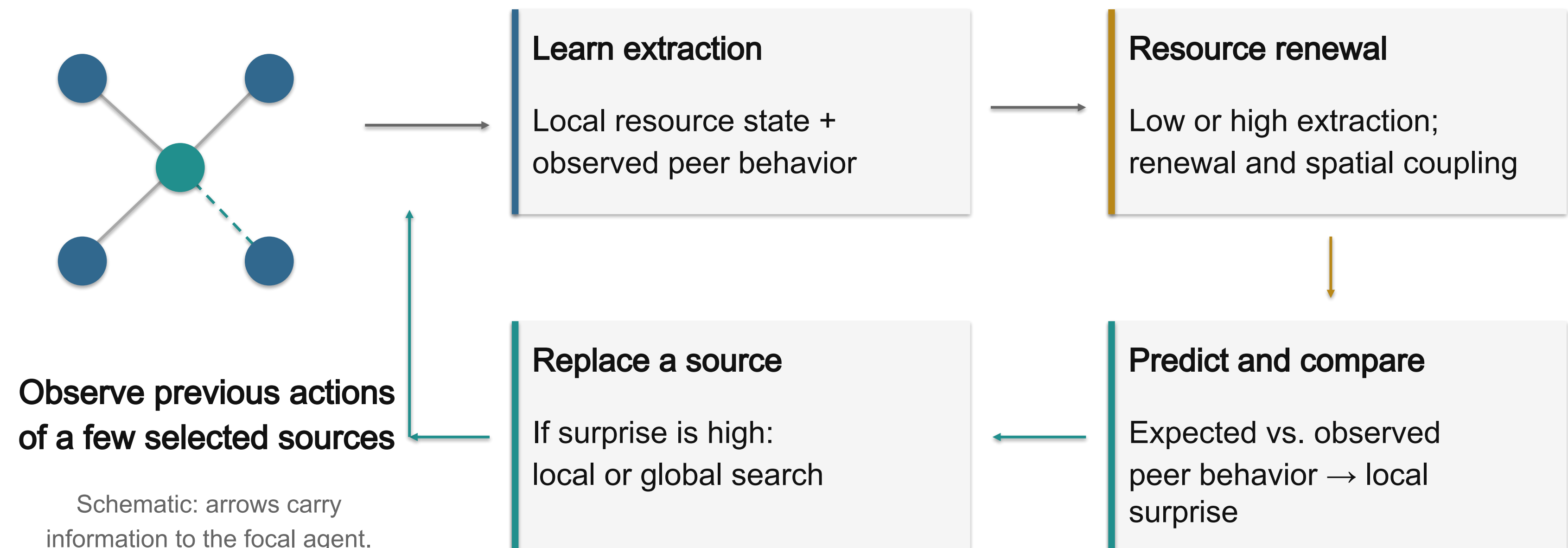
FROZEN HYPOTHESES

H1: perception varies with ecology. H2: local search concentrates visibility.

H3: adaptive selection changes perception relative to matched random turnover.

Population information feeds decisions about local resources

Agents stay in place; information links change. Population accuracy may not imply local usefulness.



Reward = own harvest. Welfare is measured separately; it does not enter the reward.

The full run separates ecology, search scope and observer selection

27 conditions × 100 replicates = 2,700 condition-replicates · 64 agents per population

Three reference conditions

- B0** Ecological learning only
- S1** Fixed random observation network
- S2** Uneven visibility + varying attention

R1 / R2 / R3 ↔ matched R0

Local / mixed / global search
Same rewiring event counts and scope
Different selection of observers



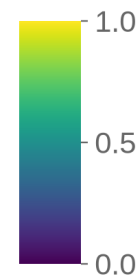
Uniform high



Patchy high



Split high / low

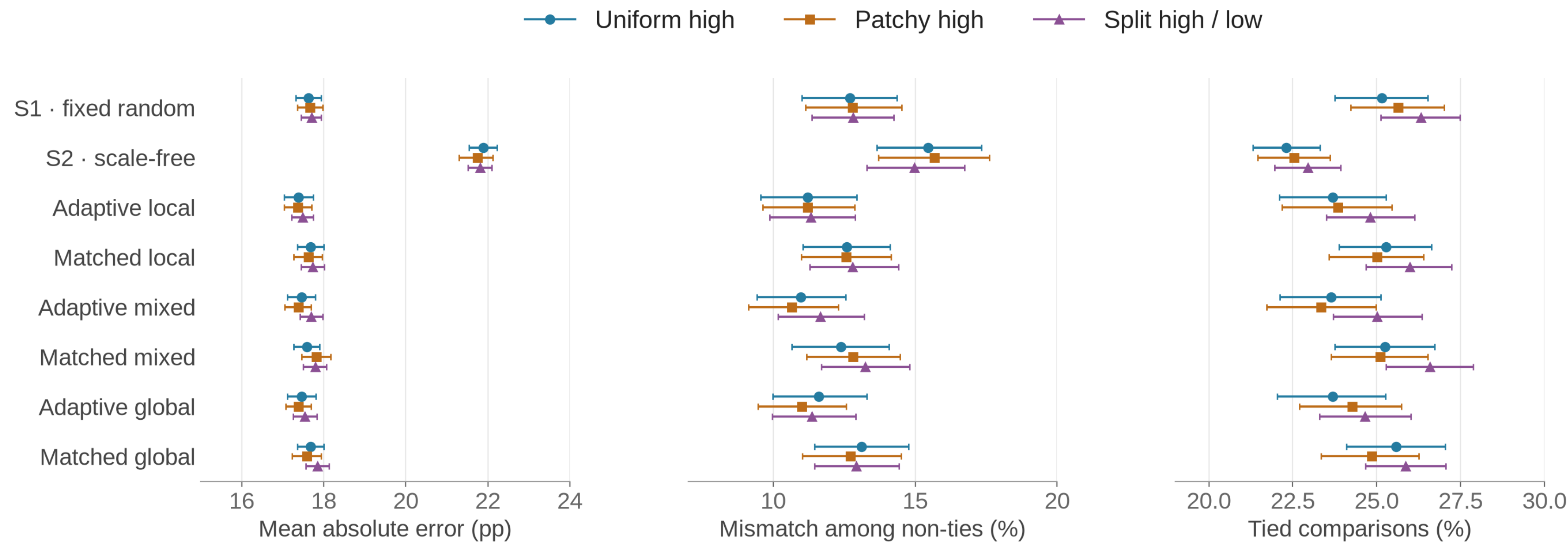


3 ecologies × 9 observation rules
100 replicates per condition
5,000 training + 1,000 evaluation steps
Outcomes: fresh reset, frozen policies + graphs

Illustrative capacity maps · common scale 0–1

Observation rules change perception; ecology differences are small

Final 1,000 training steps · all eight social treatments · means and 95% bootstrap intervals

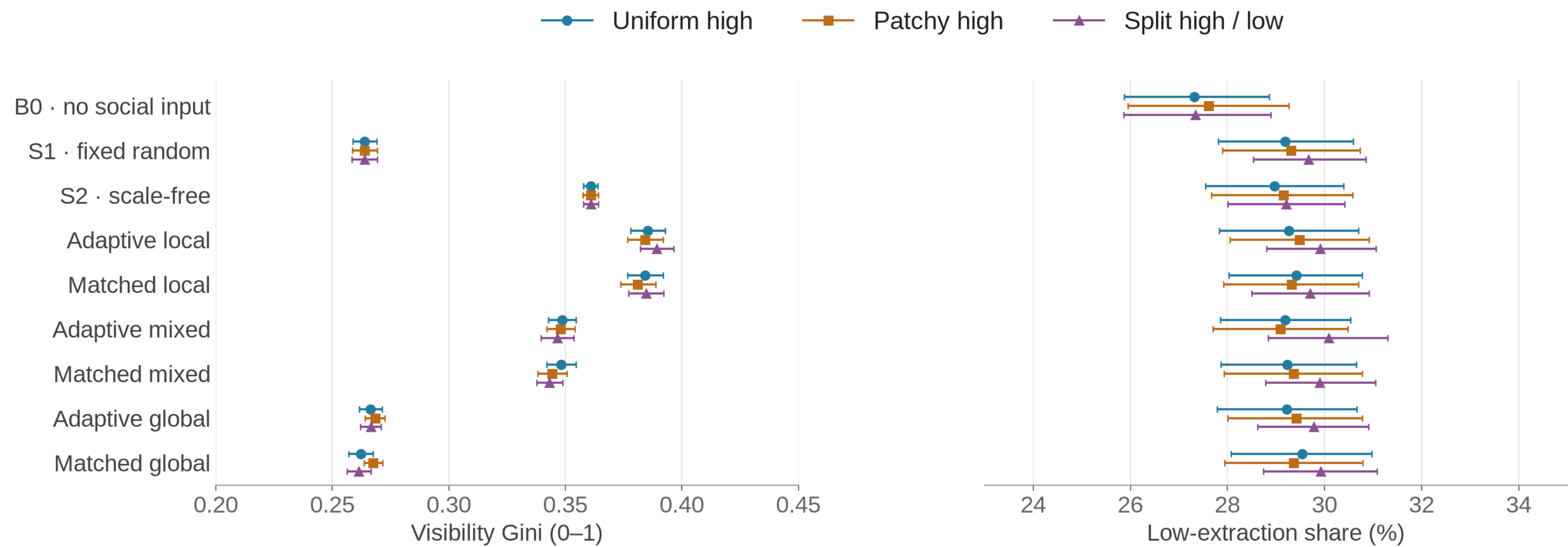


Adaptive selection lowers non-tie mismatch by 1.35–2.16 pp versus matched turnover.

Raw error includes sampling: four random peers already imply 17.5–17.8 pp MAE. S2 also varies attention capacity.

Local search concentrates visibility, even under random turnover

Collective organization = visibility concentration and prevalence of low-extraction behavior

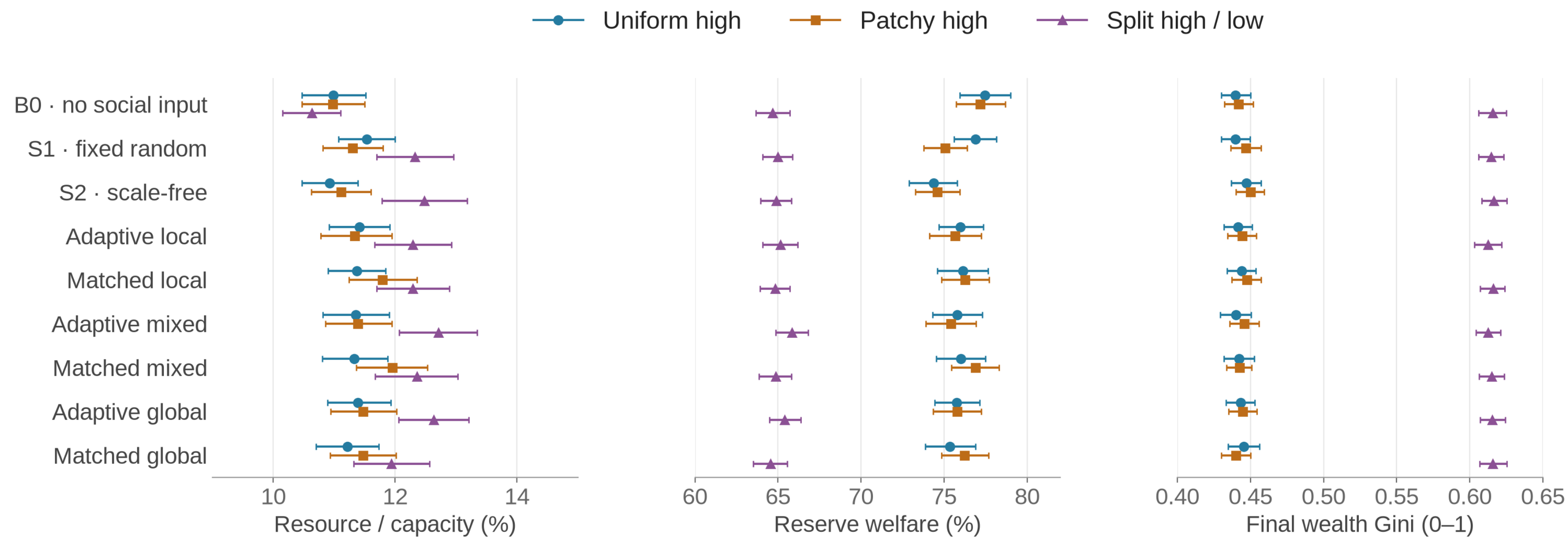


Local versus global search: +0.116–0.123 visibility Gini under adaptive rewiring.

Matched random turnover produces a similar gap. All nine adaptive–matched extraction CIs include zero.
100 replicates; final 1,000 training steps; pointwise 95% bootstrap CIs. “Local” means social-graph proximity.

Ecology changes welfare and inequality; adaptive benefits are inconsistent

Absolute outcomes · 1,000-step fresh-reset evaluation · frozen policies and terminal networks



Split versus uniform: welfare –9.5 to –11.9 pp; wealth Gini +0.169 to +0.175.

Adaptive–matched welfare: only 1 of 9 pointwise CIs excludes zero. Split changes several ecological parameters together.
Reset matters: mean reserve welfare is 72% after reset versus 26% in continuation. This is not long-run sustainability.

The next question is when information becomes useful for decisions

Completed study: ecology → population perception and organization → material consequences

What the run establishes

Observation rules shift perception and concentration.

Ecology shifts welfare and wealth inequality.

What remains unresolved

Better population estimates do not consistently improve

local decisions or welfare.
The causal link is untested.

One bounded next study

One Harvest environment; fixed learner and attention.

Informative versus shuffled signals; matched productivity.

Feedback I want from you

Are RQ1 and RQ2 clear, and do these measures capture collective organization?

Should the next hypothesis target decision usefulness rather than global accuracy?

Would SocialJax Commons Harvest justify migration, or should we retain this model?